

EE 372 Random Signals

Assignment #4

Due date/time: April 27, 2025 23:59

$X(t)$  is be a WSS process with mean  $\mu_X = 0$  and autocorrelation  $R_X(\tau)$ . For the following parts, determine whether  $Y(t)$  is WSS or not by first calculating its mean and autocorrelation. Show your calculations.

- $Y(t) = X(-t)$
- $Y(t) = aX(t)$
- $Y(t) = X(t^2)$
- $Y(t) = X(t)\cos(\omega) + X(t - \tau_0)\sin(\omega)$

For a process to be wide-sense-stationary (WSS)

- $E[X(t)] = \mu_X$
  - $R_X(t, \tau) = E[X(t)X(t+\tau)] = R_X(\tau)$
- } independent of time  $t$

a)  $Y(t) = X(-t)$

$$E[Y(t)] = E[X(-t)] = 0 \quad \text{since } X(t) \text{ has zero mean at any } t_0 \in [-\infty, \infty]$$

$$R_Y(t, \tau) = E[Y(t)Y(t+\tau)] = E[X(-t)X(-t-\tau)] = E[X(-t-\tau)X(-t)] = R_X(-t, \tau) = R_X(\tau)$$

↳ Thus,  $Y(t)$  is WSS as well

b)  $Y(t) = aX(t)$

$$E[Y(t)] = E[aX(t)] = aE[X(t)] = a \cdot 0 = 0$$

$$R_Y(t, \tau) = E[aX(t) \cdot aX(t+\tau)] = a^2 E[X(t)X(t+\tau)] = a^2 R_X(\tau)$$

↳ Thus,  $Y(t)$  is WSS

c)  $Y(t) = X(t^2)$

$$E[Y(t)] = E[X(t^2)] = \mu_X = 0 \quad \forall t_1, \text{ there exist } \exists t_2 = t_1^2 \text{ such that } E[Y(t_1)] = E[X(t_2)] = \mu_X \text{ since } X \text{ is WSS}$$

$$R_Y(t, \tau) = E[X(t^2)X((t+\tau)^2)] = R_X(t^2, (t+\tau)^2 - t^2)$$

$$\text{Define } t_0 = t^2 \Rightarrow R_Y(t, \tau) = E[X(t_0)X(t_0 + 2\tau\sqrt{t_0} + \tau^2)] = R_X(t_0, 2\tau\sqrt{t_0} + \tau^2) = R_X(\tau^2 + 2\tau\sqrt{t_0})$$

↳ This time  $R_Y(t, \tau) = R_X(\tau^2 + 2\tau\sqrt{t_0})$  is not solely a function of  $\tau$ . It also depends on  $t_0$  and thus  $t$ . For example  $R_Y(2, 1) = R_X(4, 5) = R_X(5)$  while  $R_Y(3, 1) = R_X(9, 7) = R_X(7)$ .

↳ Thus  $Y(t)$  is NOT WSS.

d)  $Y(t) = X(t)\cos\omega + X(t-\tau_0)\sin\omega$

$$E[Y(t)] = E[X(t)\cos\omega] + E[X(t-\tau_0)\sin\omega] = \cos\omega \cdot \mu_X + \sin\omega \cdot \overbrace{E[X(t-\tau_0)]}^{\text{also } \mu_X \text{ since WSS}} = (\cos\omega + \sin\omega) \cdot 0 = 0$$

$$\begin{aligned} R_Y(t, \tau) &= E[(\cos\omega X(t) + \sin\omega X(t-\tau_0))(\cos\omega X(t+\tau) + \sin\omega X(t-\tau_0+\tau))] \\ &= \cos^2\omega E[X(t)X(t+\tau)] + \sin^2\omega E[X(t-\tau_0)X(t-\tau_0+\tau)] \\ &\quad + \sin\omega\cos\omega E[X(t)X(t-\tau_0+\tau)] + \sin\omega\cos\omega E[X(t-\tau_0)X(t+\tau)] \\ &= \cos^2\omega R_X(\tau) + \sin^2\omega R_X(\tau) + \sin\omega\cos\omega \{ \underbrace{R_X(t, \tau-\tau_0)}_{R_X(\tau-\tau_0)} + \underbrace{R_X(t-\tau_0, \tau+\tau_0)}_{R_X(\tau+\tau_0)} \} \end{aligned}$$

since WSS.

$$R_Y(t, \tau) = (\cos^2\omega + \sin^2\omega) R_X(\tau) + \sin\omega\cos\omega [R_X(\tau-\tau_0) + R_X(\tau+\tau_0)]$$

→ Noting  $\tau_0$  and  $\omega$  are mere constants,  $R_Y(t, \tau)$  is independent of  $t$ .

↳ Thus,  $Y(t)$  is WSS